Yaoyuan Zhou

My Website amiao1120@gmail.com GitHub

Education

Beijing Normal University *Sept 2023 – Jul 2027*
BS in Educational Technology (major)
- GPA: 3.7/4.0 (Average Score: 87.27)
- **Coursework:** Educational Technology (89), Introduction to Educational Research (93), Principles of Computer Composition (92), Educational Measurement and Evaluation (92), learning science (92)

BS in Artificial Intelligence (minor)
- **Coursework:** Algorithm Design and Analysis, Foundations of Program Design, Object-oriented Program

Hong Kong Polytechnic University *Sept 2025 – Jan 2026*
Exchange Student
BSc(Hons) in Artificial Intelligence and Information Engineering
- **Coursework:** Foundation Techniques in Artificial Intelligence, Artificial Intelligence Enabled Internet of Things, Network Security

Publications

Journal

An Active Instructional Approach Based on SAMR Framework to Assist Undergraduate Freshmen in Integrating AIGC into Learning: An Exploratory
The Internet and Higher Education**(SSCI Q1, Third Author)** Link
Juan Wu*, Jingwen Pan, ***Yaoyuan Zhou***, Mengyu Liu, Yanling Li, Ronghuai Huang,

Exploring the Effects of Generative AI Prompting Instruction on Undergraduate Students' Creative Problem Solving: From Learning to Using in a Classroom-Based Social Experiment
Smart Learning Environment (**SCI Q1, Third Author**,Under Review)
Juan Wu*, Jingwen Pan, ***Yaoyuan Zhou***, Mengyu Liu, Ronghuai Huang

Exploring the Second-Level Digital Divide in Elementary Science Classrooms: A Video-Based Comparative Analysis
International Journal of Science and Mathematics Education (**SSCI Q1, Second Author**,Under Review)
Yuze He, ***Yaoyuan Zhou***, Juan Lai, Fengrui Xie, Yonghe Zheng, Jie Yang*

Conference

Exploring the Pattern and Influence Factors in the Correlation Between Learning Engagement and Academic Achievement in Online Learning
Global Chinese Conference on Computers in Education, GCCCE 2026 **(First Author)**
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

Power of Query——Analysis of Critical Thinking Trajectories Based GAI-Assisted Learning
Global Chinese Conference on Computers in Education, GCCCE 2025 **(First Author)** Link
Yaoyuan Zhou, Juan Wu*

A Typological Study of Learning Engagement and Academic Achievement in University Online Classes :An Epistemic Network Analysis Based on Bloom's Taxonomy
Research Association of Learning Science Conference, RALS 2025 **(First Author)**
Yaoyuan Zhou, Yuhan Gai, Qingyang Liu, Jiamin Ren

Project

Exploring the Effects of GAI Prompting on Undergraduate Creative Problem Solving: A Task-Driven Experimental Study ***Person in Charge*** *May 2024 - Dec 2025*
Municipal-level scientific research project
- Developed a classroom AI agent website tool to support real-time human–AI interaction between students and a generative AI (GAI) chatbot, leading the end-to-end system design and implementation.
- Independently designed the course structure and created hands-on prompt engineering tutorials, integrating

GAI into the curriculum to foster undergraduates' critical and creative thinking as a central research focus.
- Conducted quantitative learning analytics using Epistemic Network Analysis (ENA) to model co-occurrence and temporal links among students' cognitive moves in chatbot dialogues.

A systematic review of intelligent tutoring systems based on Generative Artificial Intelligence — ***Person in Charge***
Aug 2025 - Now
The Education University of Hong Kong - Hong Kong, China
- Conduct a systematic review and meta-analysis of empirical studies on generative artificial intelligence (GAI) in intelligent tutoring systems (ITS), including protocol design, database searching, study screening, data extraction, and quantitative synthesis.
- Leverage a solid understanding of classic ITS architectures (domain, student, tutor, and interface models) to lead in-depth analyses of how GAI integration reshapes internal module design and interaction mechanisms, identifying evidence-based pathways for GAI to enhance ITS.

Research on the Systematic Thinking of Artificial Intelligence-Assisted Reading in Chemistry Teaching for Middle School Students — ***Person in Charge***
March 2026 - Now
Beijing Normal University & Beijing No.4 High School(BHSF)
- Responsible for the entire process of proposing research questions, course design, technology development and course implementation. Collaboratively develop educational intelligent agents to assist middle school students in learning chemistry experiments and exercising systematic thinking.

Enhancing Education: A Human-Centred Generative AI Pipeline for Academic Excellence — ***Research Assistant***
Jul 2026 - Oct 2026
Mitacs Globalink Research Internship Program (University of Regina) - Canada
- Suport the design and evaluation of an LLM-based educational system that adapts content to student needs, focusing on intelligent assessment, curriculum peronalization, and real-time feedback mechanisms.
- Conducted quantitative research involving benchmarking, performance analysis, and statistical modeling to validate system efficacy, integrating findings into iterative development cycles.

Work Experience

Project Intern - The UNESCO International Research and Training Centre for Rural Education (INRULED) — *Nov 2024 – Dec 2025*
Information and Communication Technology (ICT) in Education for Rural Development Department
- Facilitated the joint SEAMEO STEM-ED and UNESCO INRULED "STEM Project" by conducting background research on rural STEM education, drafting cooperation frameworks and evaluation criteria.
- Assisted 5 large-scale international events, managing concept notes, round-table session design, foreign expert liaison, and on-site multimedia in high-pressure, multilingual settings.
- Conducted field surveys across Chongqing, Jiangsu, and other regions to assess the current implementation status of ICT education.

Project Intern - Research Institute of Science Education-BNU
Course Design department — *Oct 2023 – Jun 2025*
- Designed a 12-hour STEM curriculum rooted in Project-Based Learning (PBL) principles within the theme of aviation, developing instructional materials that foster critical thinking and practical STEM applications.
- Led collaborative efforts with 20+ frontline STEAM educators to iteratively refine course components, driving user-centered design and optimizing educational content.

Online Teacher - Teaching Digital Skills and SDGs to Primary School Students in Rural China (UNDP) — *Jan 2024 - Jan 2025*
- Served as an AI literacy instructor at the 2024 HER Digital Future Camp, delivered 20 hours of instruction to primary school students and promoting girls' participation in STEM through hybrid learning experiences.
- Developed introductory AI lessons based on prompt-driven human–AI interaction, fostering students' critical thinking, problem-solving abilities, and early self-regulated learning skills.

Special Skills

Languages: Chinese(Native Speaker), English(IELTS 7.0), Cantonese(Listen)
Programming Languages: C, Java, Python, SQL, JavaScript
Official skills: Microsoft softwares (Word, Excel, PPT, Visio, etc.), Adobe softwares (PS, PR), Xmind